Cryptocurrency Scam Trend Analysis and Temporally Robust Detection Using Machine Learning: Evidence from Solana, Ethereum, and Bitcoin Blockchain Data

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***Abstract*—**Cryptocurrency-related scams are among the costliest forms of cybercrime: the FBI Internet Crime Complaint Center (IC3) attributed more than US$11 billion in reported losses to cryptocurrency complaints in 2025, up from US$9.3 billion in 2024 and US$5.6 billion in 2023. Many machine-learning studies evaluate scam detectors on random splits, which can overstate performance because scam tactics and blockchain behavior change over time. This paper presents CryptoRadar, an offline multi-dataset framework that combines descriptive trend analysis of reported fraud losses with leakage-aware, time-ordered evaluation of separate models for Solana liquidity pools (SolRPDS), Bitcoin and Ethereum addresses (Real-CATS) and Bitcoin transactions (Elliptic). Logistic regression and random forest baselines are compared with XGBoost using validation-selected decision thresholds and a test period scored once, in a single reproducible run. On held-out temporal test sets XGBoost reached PR-AUC of 0.444 on the SolRPDS inactivity-based proxy task, 0.223 on activity-filtered Bitcoin addresses, 0.702 on Ethereum addresses and 0.790 on Elliptic transactions. PR-AUC fell by 0.16–0.25 from validation to test; falling positive rates explain much of the precision loss on SolRPDS, Bitcoin and Ethereum, whereas Elliptic shows genuine drift. SHAP and a data audit expose three artefact risks: the proxy label, token-identity features on SolRPDS, and zero-activity criminal Bitcoin addresses. Scores are uncalibrated, dataset-specific and not comparable across datasets; they are decision-support signals requiring human review, not proof of crime.

Index Terms*—cryptocurrency scams, cybercrime trend analysis, machine learning, XGBoost, SHAP, temporal validation, Solana, Ethereum, Bitcoin, class imbalance.*

I. INTRODUCTION

Cryptocurrency has become a primary payment channel for online fraud. In its 2025 report, the FBI IC3 recorded 181,565 complaints involving cryptocurrency with reported losses above US$11 billion, and identified investment fraud as the main driver of scam losses [1]. The same series stood at roughly US$9.3 billion in 2024 [2] and US$5.6 billion in 2023 [3]. Because blockchain transfers are fast, cross-border and effectively irreversible, early identification of scam-linked addresses is more valuable than post-incident recovery.

Public blockchains expose every transaction but not the identity or intent of the owner, so supervised learning on address-, pool- or transaction-level features is a natural way to flag suspicious entities [4]–[6]. A recurring weakness is evaluation design. Random splits let near-duplicates of one campaign appear in both training and test data, and scam operators adapt, so a detector must be judged on genuinely later data. A second weakness is dataset construction: labels, collection procedures and class balance can create shortcuts that a classifier exploits without learning any behavioral fraud signal.

This work addresses four research questions. *RQ1:* How have reported cryptocurrency-fraud losses changed over recent years, and which scheme types drive the change? *RQ2:* How much does classifier performance fall when models trained on earlier periods are tested on later ones, and how much of that fall is explained by a shift in class prevalence rather than by changed behavior? *RQ3:* Which features drive predictions, and are any of them artefacts of dataset construction? *RQ4:* Do the conclusions hold across four datasets on three blockchains, and do design choices such as ablations and zero-activity rows change them?

The contributions are: (1) a trend analysis of crypto-fraud losses from official IC3 reports; (2) a documented, leakage-aware, time-ordered pipeline applied to seven model variants on four datasets; (3) a decomposition of the validation-to-test precision drop into prevalence shift and drift; and (4) an audit of three dataset artefacts with explicit limits on what the results do and do not show.

II. RELATED WORK

Weber et al. released the Elliptic transaction graph of 203,769 Bitcoin transactions, of which 4,545 are labelled illicit and 42,019 licit, and evaluated graph convolutional networks and Random Forests for anti-money-laundering [5]. Elmougy and Liu extended it as Elliptic++ with 822k

wallet addresses and four graph views [6]. For Ethereum, Wu et al. detected phishing accounts using network embeddings [7], and Li et al. proposed SIEGE, a self-supervised incremental graph-learning method [8]. Shi et al. introduced Real-CATS, a real-world dataset of criminal and benign addresses with transaction profiles for several chains [4]. Alhaidari et al. released SolRPDS, the first public rug-pull dataset for Solana, built from 3.69 billion transactions [9].

Graph models capture multi-hop structure but need transaction edges and are costlier to train. Tabular profile models are simpler and easier to explain, so they are a reasonable baseline. Our contribution is not a new architecture but a stricter, multi-dataset evaluation of tabular baselines under temporal ordering with explicit leakage controls and artefact audits.

III. DATASETS

A. Official loss statistics (RQ1). Annual loss and complaint figures were taken from the IC3 reports for 2023–2025 [1]–[3]. They are victim-reported and U.S.-centric, so they understate global losses and carry reporting bias; we use them only to describe direction and scheme composition.

B. SolRPDS. Four yearly files (2021, 2022, 2023, Jan–Nov 2024) were concatenated into 116,304 pool–mint rows (3 exact duplicates dropped; 63,520 unique pools; 1,703, 3,695, 15,477 and 95,433 rows per file, so most rows are from 2024). SolRPDS provides no independently verified rug-pull label. The target is the inactivity-based proxy y = (INACTIVITY_STATUS = Inactive), positive for 22,555 rows and negative for 93,749. A pool can be inactive for benign reasons, and a fraudulent pool can still be recorded as active, so all SolRPDS results concern this proxy only.

C. Real-CATS (Bitcoin and Ethereum). Real-CATS provides address-level transaction profiles for criminal addresses from public reports and benign addresses from exchange customers [4]. The Bitcoin files contain 90,597 criminal and 90,164 benign addresses; the Ethereum files contain 12,572 criminal and 16,019 benign addresses. Monetary units were converted (satoshi→BTC, Wei→ETH) and sanity-checked against the dataset's own USD columns (median implied prices of 35,814 USD/BTC and 1,883 USD/ETH). A key audit finding is that 55.8% of criminal Bitcoin rows (50,565) have zero observed transactions and no timestamps, versus 4.0% of benign rows (3,581); for Ethereum the shares are 13.2% and 2.6%.

D. Elliptic. The original Elliptic graph [5] has 203,769 transactions, 234,355 edges and time steps 1–49, with 4,545 illicit, 42,019 licit and 157,205 unknown transactions. Unknown transactions are excluded from training, validation and testing. This is the transaction-level dataset, not the address-level Elliptic++ extension [6].

TABLE I. DATASETS USED

Dataset	Chain	Entity	Labels	Role
IC3 reports	—	Annual statistics	Reported losses	Trend analysis
SolRPDS	Solana	Liquidity pool/token (116,304 rows)	Inactivity-based proxy	Rug-pull-like risk
Real-CATS	Bitcoin	Address (180,761)	Criminal/benign	Address risk
Real-CATS	Ethereum	Address (28,591)	Criminal/benign	Address risk
Elliptic	Bitcoin	Transaction (203,769)	Illicit/licit/unknown	Transaction risk

Elliptic: 4,545 illicit and 42,019 licit transactions are labelled; 157,205 unknown transactions are excluded from modelling.

IV. PROPOSED METHODOLOGY

Figure 1 summarizes the CryptoRadar pipeline. Each dataset has its own model; datasets are never merged and scores are not compared across models.

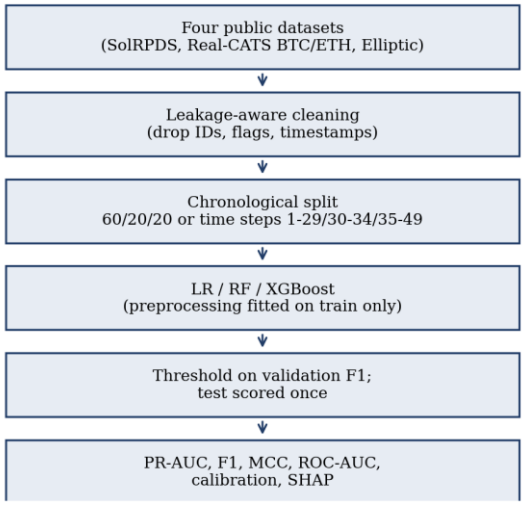


Figure 1: Proposed experimental pipeline

A. Leakage control and cleaning. Identifiers, label subtypes, collection or verification flags (watchback_checked, gs_checked, etherscan_checked), all *_transaction_id columns and absolute timestamps are never features. For SolRPDS, LAST_SWAP_TIMESTAMP, LAST_SWAP_TX_ID and everything derived from them are excluded from the primary model because the inactivity label is probably derived from activity timestamps. Median imputation (and log scaling plus standardization for logistic regression) is fitted on the training split only. The primary SolRPDS model uses 17 features, Real-CATS BTC 26, Real-CATS ETH 46 and Elliptic 165 (167 with in- and out-degree computed from the edge list).

B. Evaluation protocol. Only a time-ordered protocol was run. SolRPDS and Real-CATS rows are ordered by first-activity time and split 60/20/20 into train/validation/test; SolRPDS rows are grouped by pool address so a pool cannot straddle splits. Elliptic uses time steps 1–29, 30–34 and 35–49. Validation-selected decision thresholds (maximizing F1) are used, no probability calibration is

applied, and the test split is scored once. Test periods are SolRPDS 7 Aug–31 Oct 2024, BTC 1 Jan–29 May 2024, ETH 18 Apr 2023–5 Aug 2024. A random-split comparison, repeated seeds and bootstrap confidence intervals were not part of this run, so the random-versus-temporal gap is not quantified here. Results come from a single reproducible training run.

C. Models. Logistic regression (LR), random forest (RF) [10] and XGBoost [11] are fitted on identical splits with class weighting for imbalance and fixed default hyper-parameters; XGBoost is the main model. Implementation used Python 3.14.6, scikit-learn 1.9.1 [12], XGBoost 3.4.1 and SHAP 0.53.0 (training run of 15 Oct 2026). Validation data are used for early stopping and threshold selection, so validation metrics are optimistic relative to test.

D. Metrics. Because classes are imbalanced we headline PR-AUC (average precision) [15], and also report F1, precision, recall, ROC-AUC and the Matthews correlation coefficient (MCC) [13] for the positive class, plus Brier score and 10-bin expected calibration error (ECE). Accuracy is not reported. Precision = TP/(TP+FP); recall = TP/(TP+FN); F1 = 2PR/(P+R). We also report PR-AUC lift, the ratio of PR-AUC to the test-set positive rate, which is 1 for a random ranker.

E. Explainability. SHAP values [14] for the XGBoost model explain which inputs move its output. They are associative, not causal, and correlated features share credit.

F. Trend analysis. Trend analysis has two separated parts: (i) year-over-year IC3 losses by scheme, and (ii) counts and shares by period inside each labelled dataset. The second describes dataset composition and must not be read as a change in real-world incident rates.

V. RESULTS AND DISCUSSION

A. Loss trend (RQ1). Table II and Figure 2 show a sharp rise in reported crypto-related losses. Total losses rose about 66% from 2023 to 2024 [2], and the 2025 figure was about 22% above 2024 [1]. Crypto investment fraud grew every year in the series, from US$2.57 billion in 2022 to US$3.96 billion in 2023 [3], US$5.8 billion in 2024 [2] and US$7.2 billion in 2025 [1], about 63% of 2025 crypto losses. Complaint counts are not strictly comparable across reports because IC3 changed how crypto-related complaints are framed, so the loss series is the more reliable indicator.

TABLE II. IC3 CRYPTO-RELATED REPORTED FIGURES

Year	Complaints	Losses ($B)	Investment ($B)
2022	—	—	2.57
2023	>69,000	5.6	3.96
2024	149,686	9.3	5.8
2025	181,565	11.37	7.2

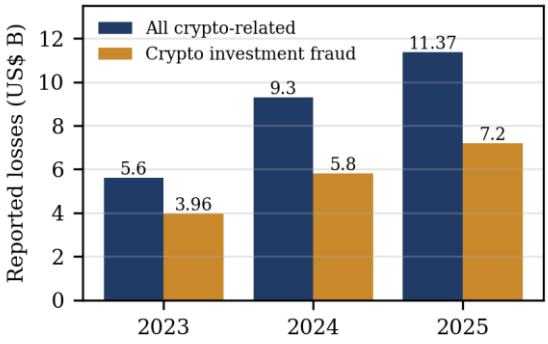


Figure 2: Reported crypto-related losses, IC3 annual reports

B. Time-ordered performance (RQ2). Table III gives the split design. The positive rate falls sharply in later periods in every dataset: from 22.4% (validation) to 10.7% (test) in SolRPDS, 7.6% to 3.6% in BTC, 40.2% to 20.0% in ETH and 16.8% to 6.5% in Elliptic. Table IV reports main-model results. XGBoost PR-AUC fell by 0.161 (ETH) to 0.245 (BTC filtered) between validation and test, and test F1 fell by 0.10–0.23, while ROC-AUC moved little (−0.085 to +0.026): ranking quality is largely preserved, but precision at a fixed threshold is not. Because validation was also used to tune the threshold and stop training, part of the gap is selection optimism.

TABLE III. CHRONOLOGICAL SPLITS (N, POSITIVE RATE)

Dataset	Train	Validation	Test
SolRPDS (primary)	69,783 (21.3%)	23,260 (22.4%)	23,261 (10.7%)
BTC (activity-filtered)	75,969 (49.0%)	25,336 (7.6%)	25,310 (3.6%)
ETH	15,904 (48.6%)	5,301 (40.2%)	5,301 (20.0%)
Elliptic	26,381 (10.9%)	3,513 (16.8%)	16,670 (6.5%)

Ablation and full/degree variants share the splits of their base dataset; BTC (full) additionally trains on 54,146 zero-activity rows (130,115 training rows, 67.4% positive).

TABLE IV. MAIN-MODEL (XGBOOST) TEST RESULTS

Variant	PR val	PR test	F1	Prec.	Recall	ROC	MCC
SolRPDS (primary)	0.681	0.444	0.418	0.301	0.685	0.844	0.355
SolRPDS (ablation)	0.581	0.421	0.484	0.370	0.701	0.867	0.429
BTC (filtered)	0.468	0.223	0.317	0.265	0.393	0.809	0.292
BTC (full)	0.437	0.208	0.297	0.227	0.429	0.810	0.277
ETH	0.863	0.702	0.634	0.539	0.768	0.888	0.535
Elliptic	0.989	0.790	0.734	0.731	0.737	0.919	0.716
Elliptic (+deg.)	0.986	0.787	0.743	0.747	0.739	0.913	0.725

Test split except PR val (validation). PR-AUC lift over test prevalence: SolRPDS 4.1×, BTC 6.1×, ETH 3.5×, Elliptic 12.2×. Single run; no confidence intervals.

These results compare model performance within each dataset. They do not imply that one model or blockchain setting is universally superior, because the datasets differ in entity type, labels, features and class prevalence; the no-skill PR-AUC equals the positive-class prevalence, so each score must be read against its own class balance.

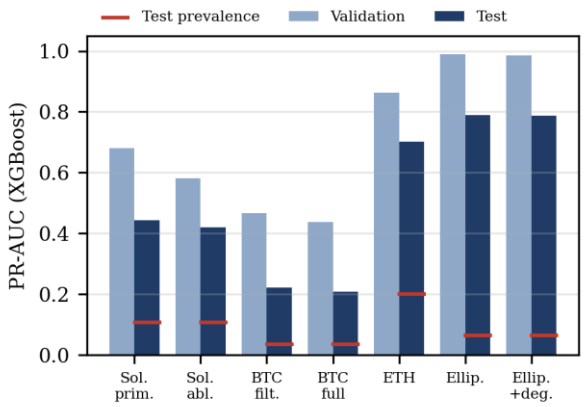


Figure 3: Validation vs. test PR-AUC; red marks show test prevalence

To separate prevalence shift from drift, Table V recomputes the precision that the validation true- and false-positive rates would give at the test prevalence. On SolRPDS, BTC and ETH, most of the precision drop is explained by the lower positive rate alone (for ETH the false-positive rate actually improved, so actual precision exceeds the expectation). Elliptic differs: recall fell from 0.975 to 0.737 and the false-positive rate doubled, so the drop reflects genuine drift, consistent with Weber et al.'s observation that models trained on early time steps struggle after a dark-market shutdown at step 43 [5].

TABLE V. PRECISION: PREVALENCE SHIFT VS. DRIFT

Dataset	Prevalence val→test	Val prec.	Expected test	Actual test
SolRPDS (primary)	22.4%→10.7%	0.543	0.331	0.301
BTC (filtered)	7.6%→3.6%	0.521	0.334	0.265
ETH	40.2%→20.0%	0.724	0.494	0.539
Elliptic	16.8%→6.5%	0.957	0.884	0.731

Expected test precision = TPR_val·π_test / (TPR_val·π_test + FPR_val·(1−π_test)).

Table VI compares algorithms on identical splits. XGBoost has the highest test PR-AUC in six of seven variants, but the margins on SolRPDS (0.444 vs. 0.435 for LR) and Elliptic (0.790 vs. 0.789 for RF) are small, and LR is higher on the SolRPDS ablation (0.492 vs. 0.421). On Elliptic, RF matches XGBoost in PR-AUC (0.789 vs. 0.790) and exceeds it in F1 (0.795 vs. 0.734) and precision (0.889 vs. 0.731), while LR collapses from a validation PR-AUC of 0.801 to 0.246. Whether ensembles degrade less than the linear baseline is therefore dataset-dependent: clearly true on Elliptic, not on SolRPDS or ETH, where all three models lose a similar 0.16–0.24 PR-AUC. With one run per configuration, differences of about 0.01 are not interpreted.

TABLE VI. ALGORITHM COMPARISON (TEST SPLIT)

Variant	PR LR	PR RF	PR XGB	F1 LR	F1 RF	F1 XGB
SolRPDS (primary)	0.435	0.396	0.444	0.310	0.354	0.418
SolRPDS (ablation)	0.492	0.355	0.421	0.385	0.443	0.484
BTC (filtered)	0.111	0.143	0.223	0.176	0.157	0.317
BTC (full)	0.114	0.138	0.208	0.165	0.174	0.297
ETH	0.606	0.608	0.702	0.518	0.546	0.634
Elliptic	0.246	0.789	0.790	0.367	0.795	0.734
Elliptic (+deg.)	0.251	0.786	0.787	0.360	0.737	0.743

Ablations. Adding timestamp-derived duration features to SolRPDS raised test F1 (0.418→0.484), MCC (0.355→0.429) and ROC-AUC (0.844→0.867) but lowered test PR-AUC (0.444→0.421) and validation PR-AUC (0.681→0.581). The evidence is mixed, so we cannot show that these fields leak the outcome, but because inactivity is probably defined from activity timestamps the ablation is reported only to size their influence. Adding in- and out-degree to Elliptic changed PR-AUC by −0.003 and F1 by +0.009, within run-to-run noise.

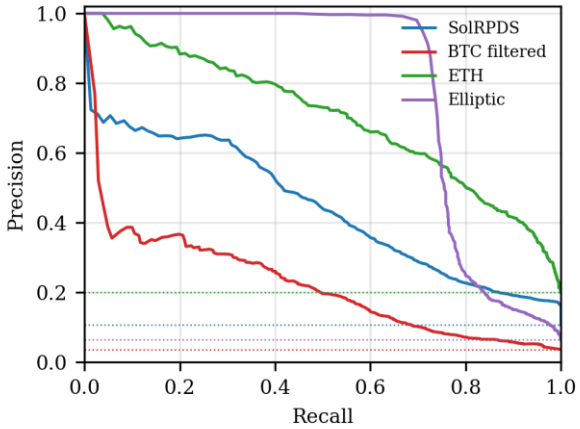


Figure 4: Test precision–recall curves (dotted: prevalence)

C. Feature importance (RQ3). Table VII lists the top SHAP features. For Bitcoin, the leading features are activity-history descriptors (lifetime, received and sent counters) followed by value-flow features (total sent, balance). For Ethereum they are value-flow (max_sent_amount, total_received_USD, max_received_amount, total_sent_USD), activity (transaction_number, lifetime) and contract interaction (from_contract_wd_normal, rank 7). Rankings are not stable across methods: XGBoost's native gain importance on ETH is led by from_contract_wd_normal (0.249, about 3.8× the next feature), whereas RF native importance is led by lifetime and SHAP by max_sent_amount, and gas-variance features appear in native but not SHAP top lists. Elliptic features are anonymized, so the top features (local_53, local_59, agg_32) cannot be given behavioral meaning; local_53, local_41 and agg_07 are high in both SHAP and native importance, and the added degree features do not enter the top eight.

For SolRPDS, mint_is_wsol has a mean |SHAP| of 1.59, 2.4× the next feature, and mint_is_usdc ranks third. These flag the pool's quote token rather than liquidity behavior and may encode how and when pools were created or labelled, which makes the 0.444 test PR-AUC hard to read

as pure behavioral signal. In the ablation, mint_is_wsol falls to 0.16 and swap_after_first_pool_hours (0.08) and swap_after_last_pool_hours (0.05) enter the top eight.

TABLE VII. TOP-5 FEATURES BY MEAN |SHAP|

Variant	Features (mean |SHAP|)
SolRPDS	mint_is_wsol 1.59; log_avg_remove_size 0.66; mint_is_usdc 0.32; num_liquidity_adds 0.20; num_liquidity_removes 0.19
SolRPDS abl.	log_avg_remove_size 0.37; mint_is_wsol 0.16; swap_after_first_pool_hours 0.08; total_events 0.07; mint_is_usdc 0.06
BTC filt.	lifetime 0.64; received_counters 0.45; sent_counters 0.40; total_sent_BTC 0.30; balance 0.27
BTC full	lifetime 0.54; received_counters 0.48; total_received_USD 0.36; sent_counters 0.32; balance 0.26
ETH	max_sent_amount 0.91; total_received_USD 0.87; max_received_amount 0.68; transaction_number 0.63; lifetime 0.60
Elliptic	local_53 1.11; local_59 0.58; agg_32 0.48; local_90 0.47; local_41 0.37
Elliptic +deg.	local_53 1.13; local_59 0.58; agg_32 0.48; local_90 0.45; agg_07 0.39

D. Cross-dataset and artefact checks (RQ4). Table IV shows the highest test PR-AUC on Elliptic (0.790, 12.2× lift), then ETH (0.702, 3.5×), SolRPDS (0.444, 4.1×) and BTC (0.223, 6.1×). These are not comparable as quality ranks: the datasets differ in chain, entity, features, labels and prevalence, and ETH's high PR-AUC partly reflects its high positive rate. The Real-CATS zero-activity audit (Table VIII) shows why the activity-filtered BTC model is the headline result. In the full model, 54,146 zero-activity rows (93.4% criminal) enter training as training-only rows, and the all-zero feature vector then scores 0.864, above the 0.686 threshold, for criminal and benign rows alike: the model can only learn "no observed activity ⇒ criminal", a property of how the dataset was assembled. These rows never occur in the dated test split, so the artefact did not help (full vs. filtered test PR-AUC 0.208 vs. 0.223, F1 0.297 vs. 0.317). Conversely, ETH zero-activity addresses, which were never seen in training, also score above the threshold (mean 0.37 vs. 0.30), so zero-activity records must be handled before scoring in any deployment.

TABLE VIII. ZERO-ACTIVITY ADDRESS DIAGNOSTIC

Model	Rows (crim./benign)	In training	Mean score	Above thr.
BTC filtered	54,146 (50,565/3,581)	No	0.522	0%
BTC full	54,146 (50,565/3,581)	Yes	0.864	100%
ETH	2,085 (1,661/424)	No	0.366/0.352	100%

All features are zero for these rows, so scores are identical for criminal and benign addresses.

Calibration. Scores are uncalibrated model probabilities and no calibration map was fitted. Test ECE is 0.028 (Elliptic), 0.059 (ETH), 0.235 (SolRPDS) and 0.370 (BTC filtered), so a score of 0.80 does not mean an 80% chance of anything; scores support ranking within one model only.

E. Error analysis. Table IX gives main-model confusion matrices. In a screening setting, false negatives are scam entities that pass unflagged and false positives are legitimate entities delayed or wrongly accused. Elliptic misses 26.3% of illicit transactions (285) with a 1.9% false-positive rate; ETH misses 23.2% (246) and flags 16.4% of benign addresses (695); SolRPDS misses 31.5% (785) and flags 19.1% of active (negative) pools (3,973), so about 2.3 false alarms accompany each true detection; BTC filtered misses 60.7% (558) with a 4.1% false-positive rate. Test prevalences here are far above real-world base rates. As an illustration, applying Elliptic's test true- and false-positive rates (0.737 and 0.019) at 1% prevalence would give precision of about 0.28, and at 0.1% about 0.04. Outputs should therefore be treated as investigative leads requiring human review, not as accusations.

TABLE IX. MAIN-MODEL CONFUSION MATRICES (TEST)

Variant	TP	FP	FN	TN	FNR	FPR
SolRPDS (primary)	1,707	3,973	785	16,796	31.5%	19.1%
SolRPDS (ablation)	1,746	2,970	746	17,799	29.9%	14.3%
BTC (filtered)	362	1,003	558	23,387	60.7%	4.1%
BTC (full)	395	1,347	525	23,043	57.1%	5.5%
ETH	814	695	246	3,546	23.2%	16.4%
Elliptic	798	293	285	15,294	26.3%	1.9%
Elliptic (+deg.)	800	271	283	15,316	26.1%	1.7%

Positive class = scam, illicit or inactive-proxy. FNR = FN/(FN+TP); FPR = FP/(FP+TN).

F. Dataset trends. Descriptive counts show that composition changes over time. SolRPDS rows are concentrated in 2024 and the monthly share labelled inactive fluctuates between near 0 and about 0.35 without a monotone trend. Criminal Bitcoin and Ethereum addresses by first-seen year peak in 2016 and 2020–2021 (Bitcoin) and in 2022 (Ethereum, phishing-dominated), and the labelled illicit share in Elliptic varies between near 0 and about 0.36 across time steps. These patterns describe how the datasets were collected and labelled and must not be read as changes in real-world incident rates; the only incident-level trend evidence in this paper is the IC3 loss series.

VI. LIMITATIONS AND ETHICS

The SolRPDS positive class is an inactivity-based proxy, not a verified rug pull. Real-CATS labels come from public reports and exchange-customer sampling, so the benign class may reflect exchange-style behavior and a classifier may partly learn that contrast. Real-CATS features summarize each address's whole history, so chronological splitting by first activity reduces but does not eliminate temporal leakage. Hyper-parameters are fixed defaults; each configuration was run once, so there are no confidence intervals or significance tests, and no random-split baseline was run; future work should add repeated seeds, confidence intervals and validation-based hyper-parameter optimization. Validation metrics are

optimistic because validation data drive early stopping and thresholds. Dataset prevalences reflect collection, not real-world base rates, and scores are uncalibrated. SHAP is associative, and artefact features can rank highly. IC3 data are voluntarily reported by U.S. victims and are not a census of global crime. Elliptic contains a large unknown-label class that is excluded from training, validation and testing, so its results apply only to the labelled subset. Only public, pre-existing datasets are used and no attempt is made to de-anonymize individuals. Predictions are investigative signals that require human review, since mislabeling an address can harm legitimate users.

VII. CONCLUSION AND FUTURE WORK

Official statistics show that reported cryptocurrency-fraud losses have grown rapidly, driven largely by investment schemes, which motivates automated screening. This paper presented CryptoRadar, a reproducible multi-dataset framework with leakage-aware preprocessing, chronological evaluation, validation-selected thresholds and SHAP explanation, applied to separate models for Solana pools, Bitcoin and Ethereum addresses, and Bitcoin transactions. Under time-ordered evaluation, XGBoost retained useful ranking ability (test ROC-AUC 0.81–0.92) with PR-AUC of 0.444 (SolRPDS proxy), 0.223 (activity-filtered Bitcoin), 0.702 (Ethereum) and 0.790 (Elliptic), but lost 0.16–0.25 PR-AUC from validation to test; much of the precision loss comes from falling prevalence on three datasets and from true drift on Elliptic. Headline numbers must be read with their artefact risks: an inactivity proxy label, token-identity features on SolRPDS, and zero-activity Bitcoin addresses. Future work includes random-versus-temporal comparisons with repeated seeds and confidence intervals, probability calibration, graph models on transaction edges, multiclass scam-type detection, verified rug-pull labels, drift monitoring with periodic retraining, live blockchain integration, and a dated incident dataset that permits a true time-series analysis of scam prevalence.

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